Congressional Votes on a Health Insurance Bill in 2009 - How did the participants vote?

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The Affortable Health Care for America Act

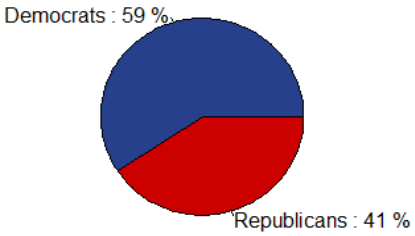
Public health care is a big point of contention in the USA between the two parties while millions of Americans struggle to afford basic health care. Republicans share the opinion that this is not a public duty and rather belongs in private hands whereas democrats see the big social benefits behind a public health care system and have tried to get it through congress several times. In 2009 under president Barack Obama, one bill succeeded: the "Affordable Health Choices Act (HR 1679)"[1]. In the following we try to find an empirical explanation for the success to see which factors led to the decision.

Data set

Therefore the data set InsuranceVote.csv will be inspected closely to provide predictions for the approval of the delegate in the House of Representatives. The data set contains nine variables and 435 observations of all eligible to vote. The variables embody information about the number of party members for republicans and democrats, which congressional district they are from, if they voted in favor of the bill or against it, information about how many citizens in their district are publicly insured, privately insured or even uninsured and lastly if the district winner in the presidential election was Obama or McCain.

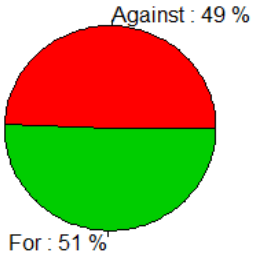
All in all, it was a tight decision for the Health Insurance Bill (215, 220). Lets take a closer look at the variables to make first assumptions about what features might have led to the vote outcome. In 2009 there were 81 democrats more in the House of Representatives (exactly 258, 177 republicans) as seen in Figure 1 below there were more democrats (59%) than republicans (41%).

Figure 1: Party Distribution in Congress



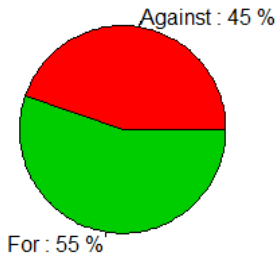
The Parties voted as following: A total of 39 Democrats (24.68% of the party) were against the bill as well as 176 Republicans (99.43%). Almost all Republicans apart from Joe Cao (R-Louisiana)2 were against the health bill while a total of 219 Democrats (75.32%) voted for the reform. Looking at these numbers a first assumption can be drawn. It seems very likely that if a representative belongs to the republican party that they vote against the bill. Although there are more democrats the result of the vote is very tight as shown in Figure 2 because 39 Democrats where against it.

Figure 2: Vote Outcome



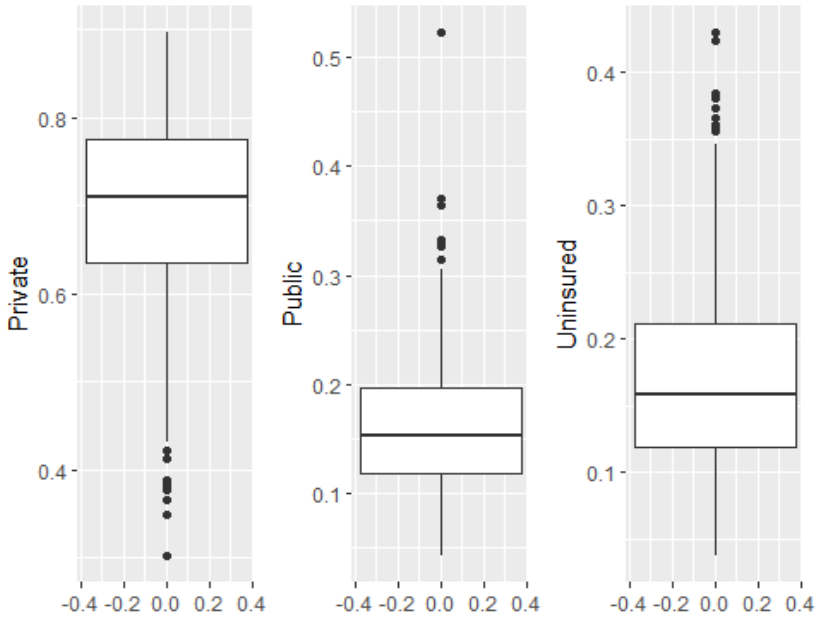
Another interesting aspect to look at is if the votes for the Bill correlate with the votes for President Barack Obama. Indeed one can find a strong positive correlation between both votes in the election 2008 and the votes of 2009 for the Health Insurance Bill of 1. As Figure 3 shows the votes for Barack Obama were 55% to 45%.

Figure 3: Percentage of Votes for Obama per district



A next thing to look closer at is what the insurance situation was like at the time of the vote since this might have an influence on the decision of the representatives' votes. On average 70% of citizens were privately insured while this ranges from 30.02% to 89.70%. Whereas on average 16.15% were publicly insured with a range of 4.3% to a maximum of 52.3% throughout the districts. Also 16.92% of citizens in each districts were on average not insured at all. This situation is also depicted in Figure 4. One could guess that a district with more uninsured citizens would have a bigger profit of the Health Insurance Reform.

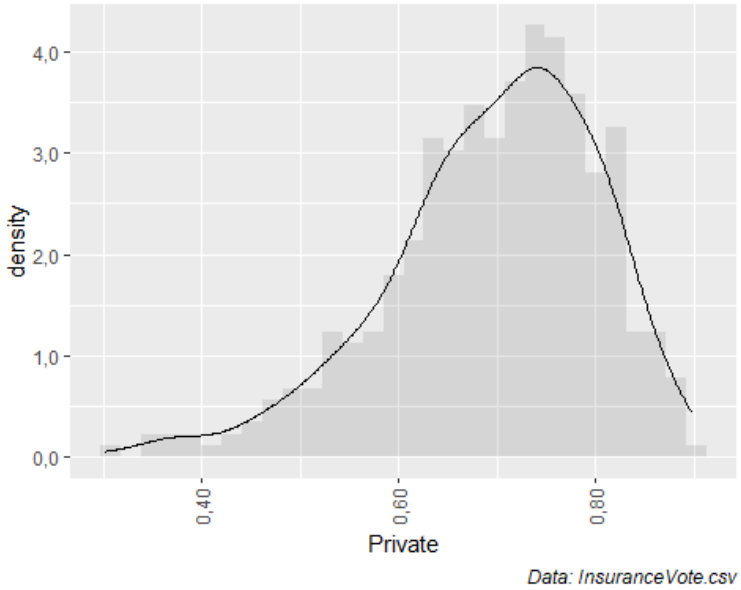
Figure 4: Comparing Private, Public and Uninsured



The following histogram in Figure 5 shows the distribution of non-senior citizens in districts with a private health insurance. This distribution is right skewed with -0.77 and could not be

improved towards the normal distribution with a log transformation. The skewness of the log transformed variable private is -1.38. A log transformation is not useful in this case.

Figure 5: Non-senior citizens in district with private health insurance



In the data set, there are two outliers replaced with the mean in the variable Public. Two districts had very high values for citizens with public insurance compared to the rest of the distribution. To make more valid predictions we will continue working with the truncated variable. The distribution of publicly insured citizens in the districts is depicted in Figure 6. Less people have a private insurance therefore the distribution is left skewed with a value of 0.83. To improve the distribution the logarithm of the variable is useful because the skewness gets closer to 0. It amounts to -0.35 for the logarithmic variable and in Figure 7 the improvement is shown. In the following investigation the truncated and logarithmic version of the variable Public is used to draw conclusions on voting for the Health Insurance Bill.

Figure 6: Percentage of non-senior citizens in district with public health insurance

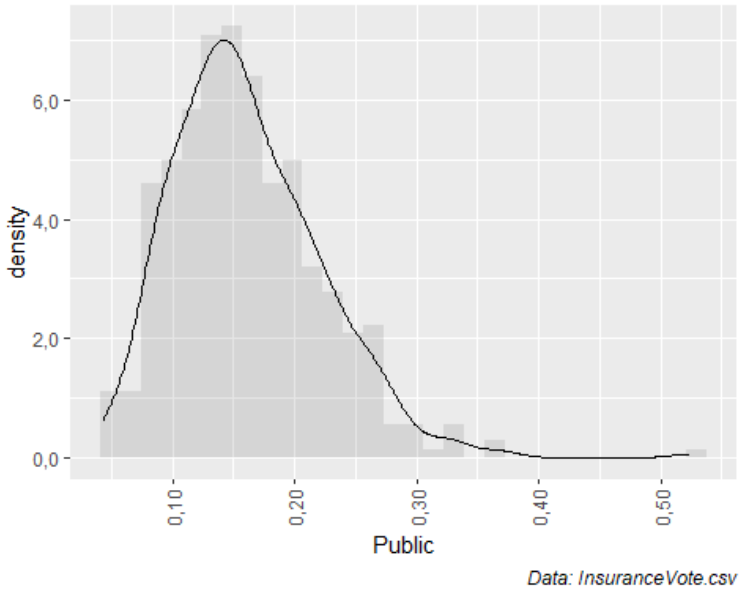
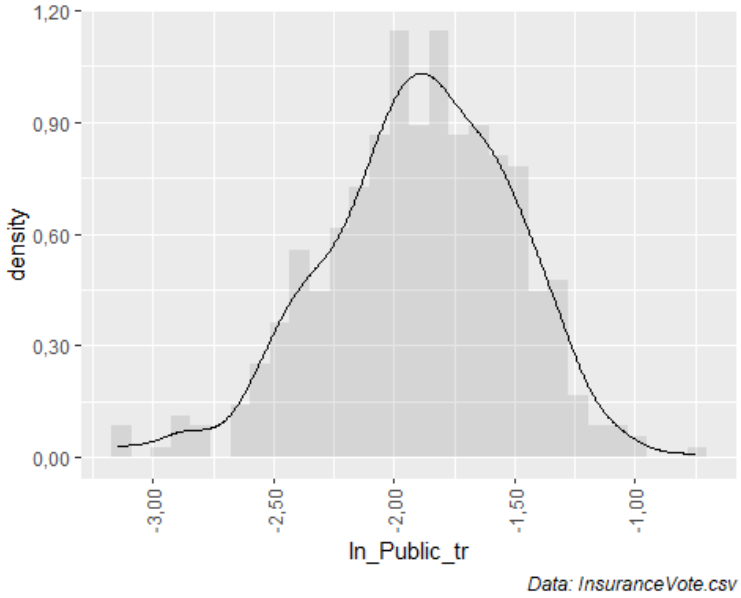


Figure 7: log transformation truncated Public



Looking at the density of uninsured citizens in Figure 8 we see again a left skewed distribution with a skewness of 0.87 and after the log transformation the improvement to -0.44. Also we will use, for the following investigation, the logarithmic Uninsured variable also depicted in Figure 9.

Figure 8: Percentage of non-senior citizens in district who are uninsured

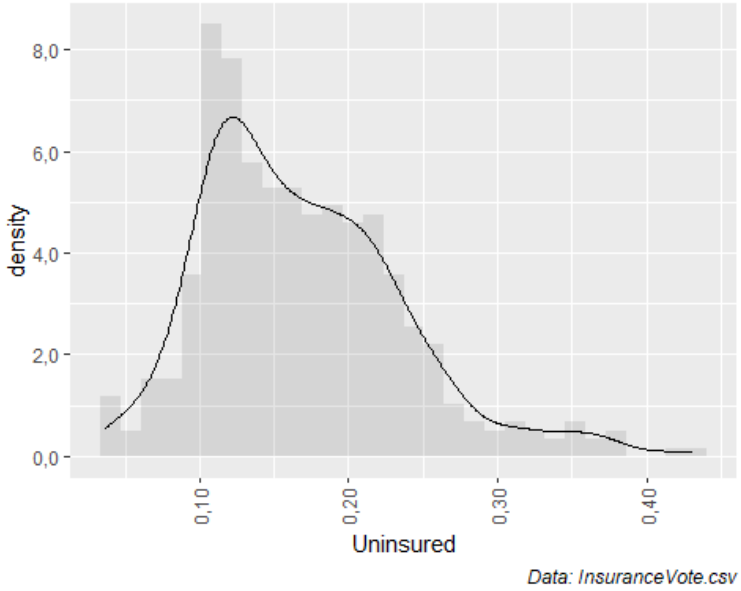
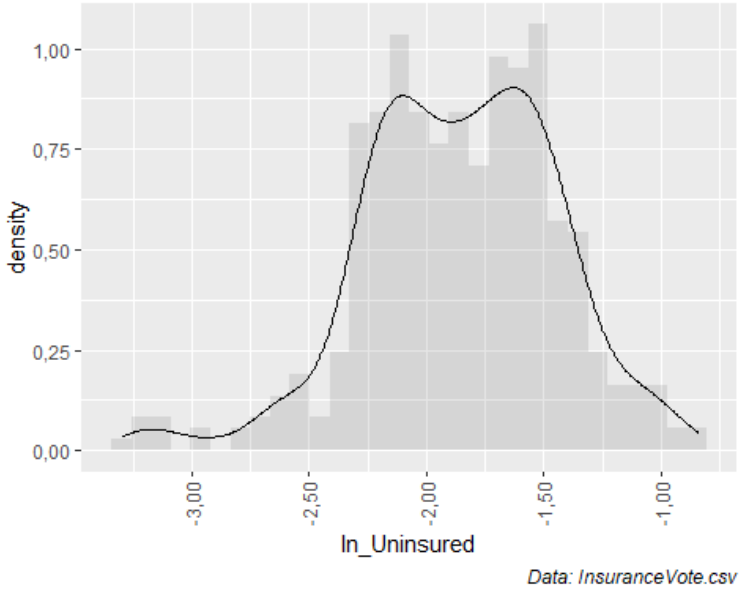


Figure 9: log transformation



As a final step after looking at descriptive statistics and data preparation, let's outline some coherence before starting the modeling process. The data set tells us that on average representatives with more privately insured citizens were less likely to vote for the Health Insurance Bill. On the other hand representatives were more likely to vote in favor if, on average, more publicly insured or uninsured people live in their district. Additionally, most members of congress who voted for Obama also voted for the health insurance bill. That information gives us some more ideas of what features can explain the votes for the Insurance Bill of 2009.

Log Models (without regularisation)

In order to start the modeling process the data set will be split into a test and training set in a ratio of 70/30. This is necessary to prevent overfitting. The model 1 is a logit model and reads as as followed Insurance Vote = beta0 + Party + Private + log(Public) + Obama + u. These predictors were chosen because they represent all informative variables we could possible include without causing colinearity. The dummy variables for Democrat and Republican were left out because they express the same information as the variable Party; the District variable was excluded because it has no informational value - it is merely an identifier; and the variable "Uninsured" was omitted because it corresponds too closely with the other percentages (seeing as the three values add up to 100%). In 93.89% of observations from the test sample this model predicted the right class of outcome which means this model is very accurate already. Additionally the Brier Score amounts to 0.22.

The Model 2 reads Insurance Vote = beta0} + Party + Private + Public + Obama + u and contains the untreated variables like they where before data processing in the data set. It shows a accuracy of 93.12% and the Brier Score of 0.22 which means it is slightly less accurate than Model 1.

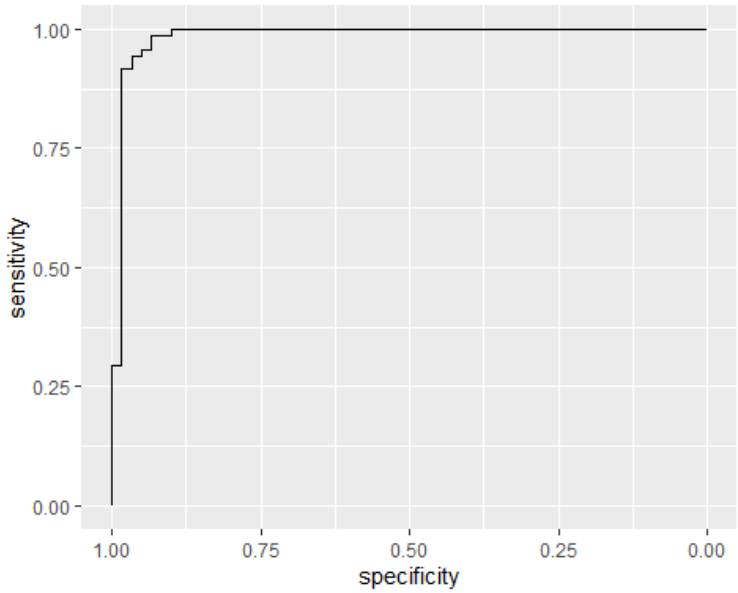
When comparing 'log1' and 'log2' it is clear that both models are equally accurate, but the model using the processed variables has a slightly lower BS. In Model 3 Insurance Vote = beta0 + log(Uninsured) + u is trying to explain the vote decision only by considering the number of uninsured citizens in the districts. The accuracy decreased to 51.90% and the Brier Score increased to 0.50 compared to Model 1 and 2. The number of uninsured people alone is not a good explanatory variable. Therefore Model 1 and 2 represent the better choice for an adequate estimation of the vote for the Health Insurance Bill.

Log Models (with Regularisation)

We applied LASSO and Ridge regression to a logistic model with the same predictors as used in 'log1'.

These AUC graph shows how capable the models are in distinguishing between the vote decisions. The graph depicted in Figure 10 corresponds to the model 'r1log' (LASSO Regression), that depicted in Figure 11 corresponds to the model 'r2log' (Ridge Regression).

Figure 10: AUC for Model 1

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## Area under the curve: 0.9847
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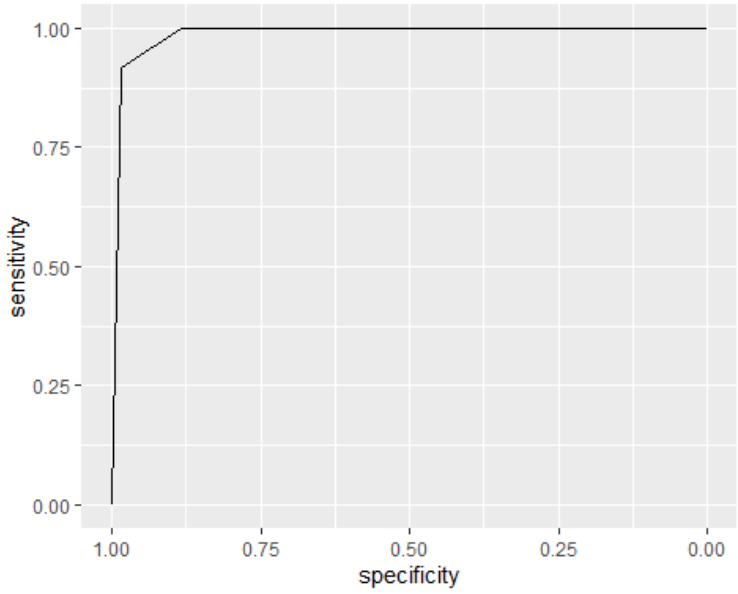
Figure 11: AUC for Model 2

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## Area under the curve: 0.9867
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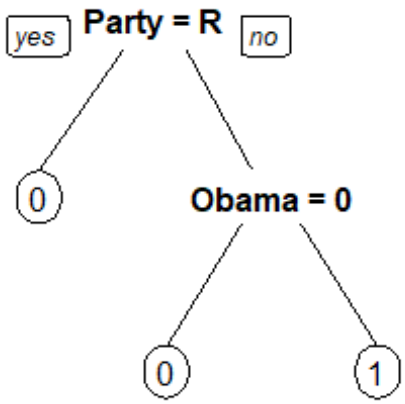
Ridge Regression offers a better model: 'r1log' has a slightly lower accuracy rate, but the AUC is closer to 1 and the BS is lower than for 'r2log'. Lastly we present a decision tree that shows that the representatives voted on average for the Insurance Bill if they belong to the democratic Party and their district voted for president Barack Obama.

Figure 12: Decision Tree



As such, the data and our modeling attempts have shown that support for health care reform is more a partisan issue (i.e. representatives are more likely to vote along party lines) than it is a reflection of the health care needs of American citizens.